power supply



Dc buck converter step down

7-12V

Custom PCB

7-12V



I2C



Arduino Uno
sensors + motors (all subsystems)

Arduino Mega
Bluetooth + LCD + buttons + game logic

PWM

5V

signal (Presence)

signal (Presence)

5V

PWM

PWM

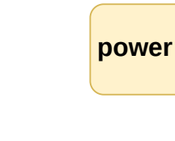
5V

signal (Presence)

Digital

I2C

serial





Servo 2 (Lifter)

IR Sensor (Presence)

IR Sensor (Presence)

L298N Motor Driver

Servo 1 (Upper)
Servo 2 (Lower)

IR Sensor (Presence)
IR Sensor (color)

Push Buttons
(White Black Reset Enter)

LCD 16 *2 I2C

HC-05 Bluetooth







DC Motor

Limit Switch

processing.io app

System 3 — Lifting (Arduino Uno + IR + Servo lifter + Servo Gate)

System 2 — Shooting (Arduino Uno + IR + DC Motor + L298N)

System1 — Sorting (Arduino Uno + 2IR + Push Buttons + LCD 16 *2 I2C +Servo 1 (Upper) Servo 2 (Lower))

communication system